

DSA STUDY BLUEPRINT SERIES

64. Next Greater Element Optimal Solution

Finding Next Greater Elements using Decreasing Stacks

Section 08 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Next Greater:** First element to the right that is strictly larger.
- Maintain index stack in decreasing element order.

Memory & Execution Blueprint

Arr: [4, 5, 2] $\Rightarrow$ Next Greater of 4 is **5**. Next Greater of 5 is -1.

C++ Implementation Reference

Standard code layout and syntax

```
vector nextGreater(vector& arr) {  
    vector ans(arr.size(), -1);  
    stack st; // indices  
    for (int i=0; i < arr.size(); i++) {  
        while (!st.empty() && arr[i] > arr[st.top()]) {  
            ans[st.top()] = arr[i]; st.pop();  
        }  
        st.push(i);  
    }  
    return ans;  
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Time Complexity	$O(N)$
Space Complexity	$O(N)$

Key Practices & Gotchas

- **Important:** Processes indices backwards if searching for next greater element to the left.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace monotonic stack modifications for next greater element on `[4, 5, 2]` :

Active Element	Stack State (Indices)	Pop comparison	Result output
4	[0] (value 4)	N/A	Waiting for next greater
5	[1] (value 5)	5 > 4 (pop index 0)	Next Greater of 4 is 5

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Monotonic properties:** Stacks process elements strictly increasing or decreasing, optimizing nested lookups.
- **Min Stack details:** Track current minimum bounds by storing elements as encoded difference values.
- **Related LeetCode Problems:** #155 Min Stack, #84 Largest Rectangle in Histogram, #42 Trapping Rain Water.